CSIRO PUBLISHING

Australian Journal of Botany, 2017, **65**, 339–347
http://dx.doi.org/10.1071/BT17011

Long-term studies of post-fire reproduction in an Australian shrubland and woodland

J. M. Harvey[A,C], *A. J. M. Hopkins*[D], *M. A. Langley*[A], *C. R. Gosper*[A,B], *M. R. Williams*[A] *and C. J. Yates*[A]

[A]Science and Conservation Division, Department of Biodiversity Conservation and Attractions, Locked Bag 104, Bentley Delivery Centre, WA 6983, Australia.
[B]CSIRO Land and Water, Private Bag 5, Wembley, WA 6913, Australia.
[C]Corresponding author. Email: Judith.Harvey@dpaw.wa.gov.au
[D]Deceased.

Abstract. Identifying appropriate fire-return times is critical for management of temperate southern Australia's fire-prone shrublands and woodlands. The time to first flowering and peak flowering are useful attributes for understanding how species and vegetation will respond to different fire intervals. Using a plant fire-response trait framework, we investigate patterns of reproduction in south-western Australian kwongan and woodland after prescribed fires with a 30-year longitudinal study, with the aim of identifying appropriate fire intervals. Modelling of post-fire flowering responses of species and aggregating responses into plant functional types showed substantial differences in the time after fire to first and peak flowering. Tests of hypotheses of flowering patterns after fire with different fire-response traits showed that (1) resprouters flower sooner than recruits of non-resprouters, (2) serotinous non-resprouters reach first and peak flowering later than do non-resprouters with soil-stored seed, (3) species in taller strata reach first and peak flowering later than those in lower strata and (4) geophytes flower earlier than other resprouters that lose their aboveground biomass in fire. The most fire interval-sensitive non-resprouting serotinous species take 15–20 years in kwongan to reach peak flowering and in *Allocasuarina* woodland 25–30 years, providing a working hypothesis for minimum fire intervals in the plant communities under investigation. Our study illustrates the value of long-term ecological studies for informing biodiversity management.

Received 11 January 2017, accepted 10 May 2017, published online 5 July 2017

Introduction

Fire is a primary agent of recurrent disturbance in the heathlands, shrublands and woodlands of temperate southern Australia. As is the case for other fire-prone regions in the world, plants in these fire-prone ecosystems possess traits that provide resilience to periodic burning (Keith *et al*. 2002*a*, 2007). Variation in traits relating to survival, reproduction, dispersal, recruitment, competitive ability and longevity contributes to a diversity of responses and makes some species vulnerable to decline under particular fire regimes (Keeley *et al*. 2011). Consequently, changes in the fire regime relative to historical variability, brought about by patterns of human settlement and land use, can result in species decline. Those species that are killed by fire and rely on seed germination after fire for population persistence are most at risk from fire-interval extremes (Bond and van Wilgen 1996). These obligate-seeding species face two threats in relation to fire interval: immaturity risk, where intervals are shorter than the time needed to establish a seed bank; and senescence risk, where intervals are longer than the life span of plants and their seed bank (Keeley *et al*. 2011; Gosper *et al*. 2013*a*).

Fire interval-related threats are particularly relevant to the floristically rich shrublands (kwongan) and woodlands of the Avon Wheatbelt Bioregion (AVW) in Mediterranean-climate south-western Western Australia (SWWA). Extensive land clearing and landscape fragmentation in the AVW have dramatically reduced the landscape continuity of native vegetation and fuels. This together with active fire suppression, contributes to substantially altered fire regimes. Parsons and Gosper (2011), in a study of contemporary fire regimes in the region, found that fire was very infrequent in small remnants (100–500 ha; estimated mean fire interval = 339 years), more frequent in large remnants (>500 ha; estimated mean fire interval = 67 years) and most frequent in uncleared areas (estimated mean fire interval = 40 years). They concluded that remnant vegetation in fragments is, on average, experiencing fire at longer intervals than would have been the case before land clearing.

There is still much to learn about the effects of very long fire intervals, relative to those experienced historically, on the dynamics of shrublands and woodlands in the fire-prone landscapes of southern Australia. Nevertheless, in the AVW, documented impacts include reduced opportunities for recruitment and increased risk of extinction in obligate-seeder (non-resprouting) species that rely on the passage of fire for

Journal compilation © CSIRO 2017

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

recruitment (Yates and Ladd 2005), senescence of obligate-seeder populations and associated declines in canopy-stored seed banks (Gosper *et al*. 2013*a*), shifts in species composition (Gosper *et al*. 2012*a*; Shackelford *et al*. 2015) and increases in abundance of inter-fire recruiting species such as *Allocasuarina huegeliana*, with associated state transitions in vegetation from kwongan to woodland (Maher *et al*. 2010). These findings have indicated that more active and nuanced fire management is necessary to conserve the region's plant diversity. However, conservation and fire managers in landscapes such as the AVW must consider when and how often it is appropriate to introduce or exclude fire in remnant vegetation.

The response of plants classified in a framework on the basis of traits relevant to their ability to persist under different fire intervals (vital attributes, *sensu* Noble and Slayter 1980) is one method of deriving ecologically appropriate fire-return intervals for plant communities. Key traits include the modes of individual or population persistence and seed storage, the timing of critical life stages, including juvenile periods and senescence, and life form (Bradstock and Kenny 2003; Gosper *et al*. 2012*a*, 2013*a*; Keith *et al*. 2002*b*, 2007). Previous studies in temperate Australian shrublands have shown that trends over time since fire in the richness and abundance of plant functional types (PFTs) based on key fire-response traits are predictable (Keith *et al*. 2007; Clarke *et al*. 2009; Gosper *et al*. 2012*a*). These studies have provided a useful model for testing hypotheses about the timing of reproduction in temperate shrublands and woodlands.

Under the vegetation-dynamics model proposed by Keith and Bradstock (1994) and Keith *et al*. (2007), and modified by Gosper *et al*. (2012*a*), serotinous non-resprouter shrubs and trees are dominant and key drivers of community dynamics through their competitive suppression, when mature, of PFTs in lower vegetation layers. Because longevity of adult plants and their seed bank through the inter-fire period is crucial for population persistence, significant resource investment is required for development of fruit structures and to attain a competitively dominant stature. It is predicted that serotinous non-resprouters will have relatively long juvenile periods, but continue flowering in longer-unburnt vegetation (Keith *et al*. 2007; Cramer and Midgley 2009). In non-resprouters with soil-stored seed, competition with dominant serotinous non-resprouters can be avoided by more rapid growth and seed-bank accumulation after fire (Keith and Bradstock 1994; Keith *et al*. 2007; Gosper *et al*. 2012*a*). Hence, we predict non-serotinous non-resprouters to have shorter juvenile periods than do serotinous non-resprouters and a decline in flowering activity with increasing time since fire. In resprouting taxa, high individual persistence is likely to ensure the presence of these taxa throughout the inter-fire period and through fires (Keith *et al*. 2007; Gosper *et al*. 2012*a*), but trade-offs in resource allocation between individual persistence (resprouting organs) and reproductive output may result in inconsistent flowering patterns (Bond and Midgley 2001). More rapid flowering after fire than non-resprouter shrubs in individuals that have resprouted (secondary juvenile period) is predicted as resprouters rapidly replace biomass lost to fire (Keith and Bradstock 1994). Because of their limited longevity and fire-stimulated germination (Keith *et al*. 2007; Gosper *et al*. 2012*a*), ephemerals are predicted to flower profusely in the immediate post-fire period and then retreat to the soil-stored seed bank.

In the present study, we describe the flowering patterns of plant species from the floristically diverse and fire-prone kwongan and woodland plant communities of the AVW in SWWA. Our study uses data from 10- and 30-year investigations of post-fire vegetation dynamics encompassing a large number of plant species across a wide spectrum of PFTs. We provide information on changes in flowering patterns to assist fire managers with defining appropriate fire intervals that minimise the 'immaturity risk' in the most sensitive species. We also test hypotheses based on fire-response traits, using the model of vegetation change with time since fire, as outlined above. Specifically, we ask the following questions:

(1) Do resprouters flower earlier and reach peak flowering earlier after fire than do non-resprouters?
(2) Do geophytes, a special class of resprouters that avoid autumn fires through retreating to belowground organs, flower earlier post-fire than do other resprouters that lose the resources in aboveground biomass in fire?
(3) Does flowering in non-resprouters with a soil-stored seed bank start earlier and decline earlier than those with a canopy-stored seed bank?
(4) Do plants of greater stature take longer to flower than those of lower stature?

Materials and methods

Study site

The study was conducted within Tutanning Nature Reserve (32°33′S, 117°20′E), located 150 km east–south-east of Perth in the western part of the cleared agricultural region in south-western Western Australia known as the wheatbelt (Fig. 1). This isolated 2140-ha remnant contains a mosaic of vegetation, including *Eucalyptus* and *Allocasuarina* woodlands, granite outcrop herbfields and shrublands, and kwongan shrublands, and is significant for the conservation of a rich diversity of flora and fauna in a heavily cleared region (Brown and Hopkins 1983).

Climate in the study region is Dry Mediterranean. Average annual rainfall (since 1891) at Pingelly 20 km west of Tutanning is 445 mm, falling mainly in the Australian winter from May to September. Summers are hot with average maximum temperatures of 31.1°C (http://www.bom.gov.au/climate/data/index.shtml accessed 22 January 2016).

The Department of Biodiversity Conservation and Attractions and its predecessor agencies have managed Tutanning since 1960. Systematic investigations of fire ecology in the reserve began with prescribed fires in 1985 and 1990, with subsequent monitoring of post-fire vegetation recovery and vegetation dynamics continuing over the past three decades.

Experimental fires

Two areas of kwongan and mixed *Allocasuarina huegeliana* and *Eucalyptus wandoo* woodland vegetation were burnt at the end of the summer dry season, in April 1985 and March 1990. The 1985 fire in a 44-ha area of kwongan and woodland (previously burnt in 1960) was of moderate intensity, hot in patches, and resulted in near-complete burning of the target area.

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

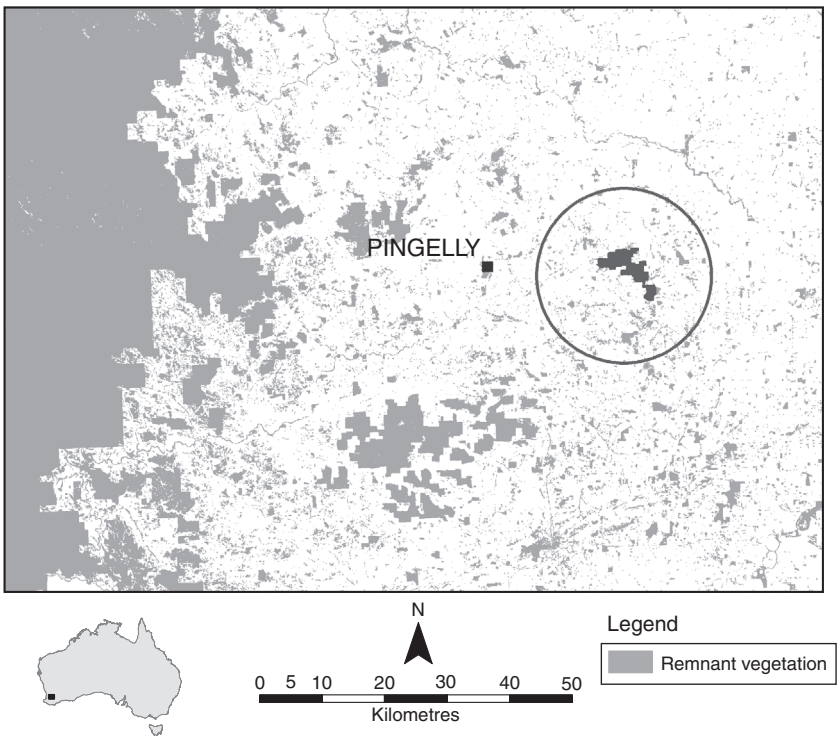


Fig. 1. Location of Tutanning Nature Reserve in south-western Western Australia.

The 1990 fire, in an area of 108 ha (previously burnt in 1965), was of moderate to high intensity. It burnt through most of the kwongan at a rate of 1100 m h^{-1}, through *Allocasuarina* woodlands in a continuous front with flame height reaching 2 m and at a rate of 200 m h^{-1}. In *Eucalyptus* woodland, the fire slowed down to 100 m h^{-1} because of lower levels of ground cover and litter and reduced exposure to wind. The canopy of the *Eucalyptus* woodland and some of the *Allocasuarina* woodland remained unburnt; there were occasional unburnt mature *Allocasuarina* plants in areas where the fire had been of low intensity, but the shrub and herb layers were mostly or completely burnt.

Sampling strategy

To understand post-fire vegetation recovery and dynamics, 21 100-m^2 circular plots (14 in woodland and seven in kwongan) were established in the spring of 1983, before the 1985 autumn fire, and resampled in the spring of 1985, 1986, 1987, 1988, 1990, 1992, 1995, 2000 and 2015. Similarly, 45 100-m^2 circular plots (29 in woodland and 16 in kwongan) established in the spring of 1988, before the 1990 autumn fire, were resampled in the spring of 1990, 1991, 1992, 1994, 1997 and 2000. Within each plot, the fire-response (resprouter versus non-resprouter) of each species was determined together with the reproductive status of post-fire recruits for non-resprouters and post-fire resprouts for resprouters.

Plant functional-type classification

We classified perennial species into PFTs on the basis of traits relevant to a vegetation assembly model putatively operating in our vegetation associations (Table 1). This model was based on the initial floristic composition model of Egler (1954), as applied to Mediterranean-climate shrublands by Keith *et al*. (2007) and Gosper *et al*. (2012*a*). The traits used were (1) the capacity of individuals to resprout from fire-resistant organs (for example lignotubers, rhizomes), (2) the location (i.e. canopy or soil) and persistence of the seedbank, (3) competitive stratum (tree, shrub and low), largely reflecting maximum plant height and (4) longevity of standing plants (i.e. species divided into those that grow, reproduce and senesce primarily in the intermediate post-fire period ($\leq$6 years; post-fire ephemerals) and those that do not). Even though responses are likely to vary among taxa classified as having a particular trait state, for example, in the proportion of a pre-fire population that resprouts after fire among resprouting taxa, this classification has proved suitable for predicting and explaining vegetation response to fire (Keith *et al*. 2007; Gosper *et al*. 2012*a*).

Not all the resultant PFT combinations were present in the sampled flora; for example, no non-resprouting taxa had transient seed banks. Further, following Keith *et al*. (2007) and Gosper *et al*. (2012*a*), we combined some allied PFTs to increase sample sizes to detect differences in reproductive behaviour, leaving nine PFTs used in the analysis (Table 1). We also included an additional PFT, cf. Gosper *et al*. (2012*a*); geophytes, a special class of resprouters that avoid hot and dry Mediterranean-type climate summers and autumns by retreating to subterranean storage organs (e.g. tubers and corms). These taxa are dormant when fires are most likely and avoided biomass loss in the study fires. Short-lived herbaceous perennial species were combined into a single PFT (ephemerals). These species occur across the lower competitive strata, and are functionally equivalent in avoiding competition with other PFTs through

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

Table 1. Plant functional types based on key traits

Traits used were longevity, individual persistence, seed-bank persistence and type and plant height (after Gosper *et al*. 2012*a*). Dashes indicate PFTs not represented in the sampled flora. PFTs were NStree, non-resprouting (N) serotinous (S) trees; NSshrub, non-resprouting (N) serotinous (S) shrubs; RSshrub, resprouting (R) serotinous (S) shrubs; NNshrub, non-resprouting (N) non-serotinous (N) shrubs; RNshrub, resprouting (R) non-serotinous (N) shrubs; NNlow, non-resprouting (N) non-serotinous (N) sedges and perennial herbs; RNlow, resprouting (R) non-serotinous (N) sedges and perennial herbs; ephem, ephemerals; geo, geophytes

Longevity	Individual persistence	Seed-bank persistence and type			Height
		Persistent, canopy (S)	Persistent, soil (N)	Transient, soil	
Long (>6 years)	Resprouter (R)	–	–		Tree
		RSshrub	[RNshrub[A]]		Shrub
		–	[RNlow/Geo[A]]		Low
	Non-resprouter (N)	NStree	–	–	Tree
		NSshrub	NNshrub	–	Shrub
		–	NNlow	–	Low
Short (<6 years)	Resprouter	–	[Ephem[B]]		Shrub
	Non-resprouter	–			Low

[A]For the purposes of our analyses, these PFTs were merged across persistent and transient soil seed-bank types.
[B]For the purposes of our analyses, these PFTs were merged across seed-bank type, individual persistence and competitive hierarchy (see Materials and methods).

Table 2. Quadrat numbers searched per post-fire age

Time since fire (years)	0.5	1.5	2.5	3.5	4.5	5.5	7.5	10.5	15.5	30.5
No. of plots	66	66	66	21	45	21	66	66	21	21

rapid growth and reproduction post-fire, then retreating belowground (in seeds or dormant tubers or rhizomes) through the bulk of the inter-fire period (see Keith *et al*. 2007; Gosper *et al*. 2012*a*). For all species recorded in plots, we used field observations and published sources to allocate them to PFTs (Table S1, available as Supplementary material for this paper).

Data analysis

We combined the 30-year (post-1985 fire; F85) and 10-year (post-1990 fire; F90) longitudinal studies of vegetation recovery and dynamics into a single time series for investigating temporal trends in reproduction among species and PFTs. We did this to increase the number of 'time-since-fire' sample points in the time series so that we had greater resolution to make inferences about reproduction and fire interval. Our study is, therefore, a combination of both longitudinal and chronosequence approaches to investigating post-fire dynamics in vegetation (Whelan 1995). Data for the time-since-fire (TSF) series was available for 21 (F85), 45 (F90) or 66 (F85 and F90) plots (Table 2).

The time to first flowering (TTFF), for each species, occurred when there was a record of the species 'flowering' (flower buds, flowers or fruits) in at least one quadrat. We modelled the relationship between time since fire (T) and flowering for each species, using logistic regression with an events and trials design. For each year in the TSF series, the number of quadrats in which a species was reproductive (F) was 'events' and the quadrats where the species occurred (O) was 'trials', with the dependent variable being flowering rate (F/O). The analysis was restricted to species where there were at least eight flowering events across all TSF classes. We fitted linear and quadratic models (i.e. by regressing F/O on T and T^2). Where the estimated coefficient for the quadratic term (T^2) in the model was statistically significant ($P < 0.05$), it was concluded that the response was non-linear. If the coefficient for the linear term (T) alone was statistically significant ($P < 0.05$), the response was deemed linear. We estimated the time to peak flowering (TTPF) from examination of modelled responses (see examples in Fig. 2). In those cases where neither the linear or quadratic response models were statistically significant, we deemed that there was no distinct peak in the flowering response. We undertook all logistic regression analyses in SAS (v. 9.4, SAS Institute Inc., Cary, NC, USA).

We investigated the effect of PFT on TTFF and TTPF with one-way ANOVA and tested for differences in mean TTFF and TTPF among the nine PFTs with a *post hoc* Tukey HSD test. The effects of particular traits on TTFF and TTPF, based on our four hypotheses, were determined using unpaired Student's *t*-tests. TTFF and TTPF (years after fire) were square-root transformed to reduce skewness and better meet assumptions of normality and homoscedasticity of residuals, but estimated differences (e.g. in TTFF among PFTs) were calculated using untransformed data. The influence of vegetation type on flowering patterns was not investigated because 50% of species were shared by kwongan and woodland, their distribution was similar across PFTs and trends to TTFF were broadly similar. We undertook ANOVA and unpaired Student's *t*-tests in Statistica (v. 7.1, StatSoft Inc., Tulsa, OK, USA).

Results

We recorded 225 perennial species in post-fire plots. In total, 165 species occurred in kwongan and 123 species in woodland understorey, with 99 common to both vegetation associations.

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

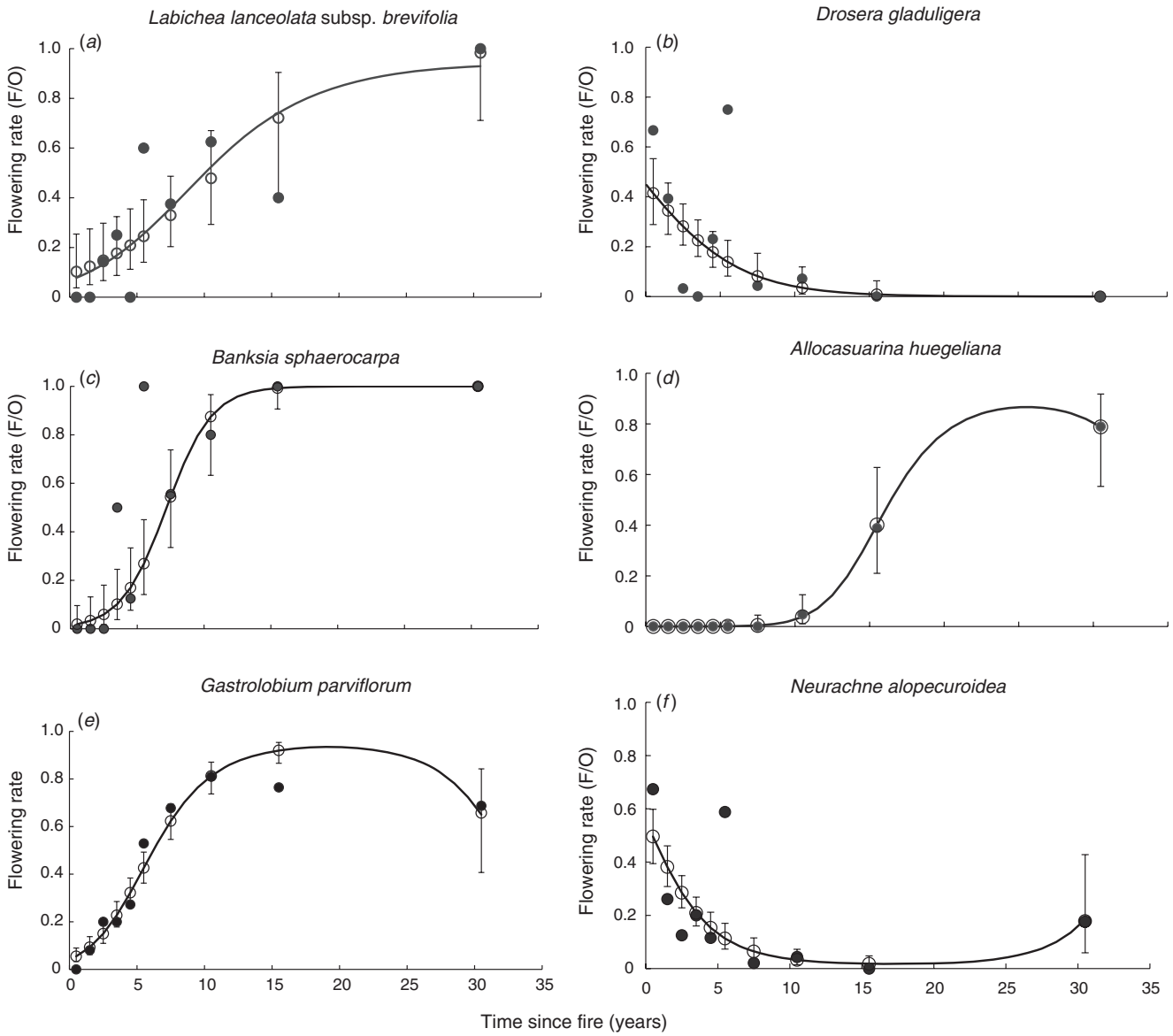


Fig. 2. Examples of flowering responses to time since fire: (*a*) ongoing increase in flowering rate after fire; (*b*) ongoing decrease in flowering rate to zero; (*c*) flowering increasing to a horizontal asymptote; (*d*) long juvenile period before a rapid increase in flowering rate; (*e*) humped response with a peak at ~15 years, before a decline in flowering rate in longer-unburnt vegetation; (*f*) decreasing response with peak in flowering immediately after fire and then flowering declines, with some evidence of an increase in long-unburnt vegetation (U-shaped). Observed (●), predicted (○) and modelled flowering rates (——).

Time to first and time to peak flowering

In total, 181 species in nine PFTs had sufficient flowering records to provide an indication of TTFF and 94 of these had enough data to attempt to model TTPF (Appendix S1). We observed distinct response types in TTPF (Fig. 2). Thirty-eight (40%) species showed a linear relationship (i.e. either increasing, decreasing or increasing to an asymptote); 22 (23%) species showed a quadratic relationship (i.e. showing a distinct peak or, in a small number of cases, a U-shape); and 34 (36%) species did not show a significant (n.s. in Table S1) response, indicating no relationship between TSF and reproductive activity, possibly owing to high levels of variation among years or a constant flowering rate. Just over 50% of the species had flowered by 1.5 years, 75% by 3.5 years and 98% by 7.5 years. TTFF varied from 0.5 years (the first post-fire sample) to 10.5 years and TTPF varied from 0.5 to >25 years (Table S1).

Plant functional types and flowering patterns

Plant functional type had a significant effect on post-fire flowering (Table 3), with mean TTFF and TTPF generally increasing with longevity and stature. As both flowering measures increased incrementally among PFTs, pairwise comparisons were not particularly informative in delineating statistical differences

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

Table 3. Number of perennial species per plant functional type (PFT), with average time to first flowering (TTFF) and peak flowering (TTPF)
PFTs were ordered by longevity (long-lived, ephemeral), stature (tree, shrub, low, geophyte), resprouting capacity (non-resprouter, resprouter) and seed-bank location (serotinous, non-serotinous). PFTs differed in TTFF (d.f. = 8,172; $F = 9.73$, $P < 0.0001$) and TTPF (d.f. = 8,51; $F = 14.9$, $P < 0.0001$), with pairwise differences based on the Tukey's HSD test indicated by different letters. Analyses were conducted on square-root transformed data, but untransformed means are shown here

PFT	Time to first flowering			Time to peak flowering		
	No. of species	Mean ± s.e.	Rank	No. of species	Mean ± s.e.	Rank
NStree	1	10.5ab	9	1	>25ab	9
NSshrub	10	4.5 ± 0.71ab	8	5	11.8 ± 1.59abcde	6
NNshrub	34	3.82 ± 0.44ab	7	12	9.6 ± 0.99bcde	5
RSshrub	12	3.58 ± 0.53ab	6	6	13.0 ± 1.12abcd	8
RNshrub	19	3.13 ± 0.48abc	5	10	12.1 ± 1.21abcd	7
NNlow	21	2.45 ± 0.43abcd	4	5	3.70 ± 1.39ef	2
RNlow	39	1.76 ± 0.34bcd	3	11	4.41 ± 1.05ef	4
Geo	32	1.44 ± 0.31cd	2	7	1.07 ± 0.57ef	1
Ephem	13	0.96 ± 0.39cd	1	3	3.83 ± 1.67cdef	3
Total	181			60		

among PFTs. Consequently, the following describes the rank order of PFT means.

Generally, geophytes, ephemerals and resprouting non-serotinous sedges and perennial herbs (RNlow) were the first PFTs to flower. Geophytes were common, making up 30% of the total perennial species. They showed a variable response; for example, some were not present immediately after fire and then emerged flowering several years later, such as, *Thelymitra maculata* and *Prasophyllum sargentii*, with *Leporella fimbriata* appearing in flower in the 7.5-year sample. However, many geophytes appear to peak at the same time as TTFF (0.5 years), indicating a positive response to fire, for example, *Drosera glanduligera*.

Ephemeral taxa, found only in the post-fire period, produced flowers the earliest after fire, on average, yet had a longer mean TTPF than did both geophytes and non-resprouting non-serotinous sedges and perennial herbs (NNlow). Consistent with their putative limited longevity and post-fire recruitment, the flowering and observation rate of ephemerals frequently declined with an increasing TSF.

RNlow (e.g. *Conostylis* spp., *Lepidosperma* spp., *Schoenus* spp., *Lomandra* spp. and *Patersonia* spp.) flowered earlier, on average, than did the non-resprouting non-serotinous sedges and perennial herbs (NNlow, e.g. *Stylidium* spp., *Laxmannia* spp., *Gompholobium* spp. and *Goodenia* spp.). However, they reached peak flowering later than did the NNlow species, mirroring their shrub counterparts. Several NNlow were short lived and most showed a decline in flowering and the number of observations after an initial early peak.

Of the shrubs, resprouting serotinous shrubs (RSshrub), often from the Proteaceae and Myrtaceae families (e.g. *Banksia sphaerocarpa*, *Hakea incrassata*, *H. lehmanniana*, *Isopogon teretifolius*), generally flowered earlier than did the non-resprouting serotinous shrubs (NSshrubs). However, they often take slightly longer to reach peak flowering than do the NSshrubs. Some RSshrubs, such as *Banksia sphaerocarpa*, *Hakea incrassata* and *Melaleuca subtrigona*, clearly showed continued flowering at 30 years since fire. The resprouting non-serotinous shrubs (RNshrub; e.g. *Astroloma epacridis*, *Hibbertia commutata*, *H. exasperata*, *H hemignosta* and *Verticordia grandiflora*) first flowered and reached peak flowering before the RSshubs. These also showed prolonged flowering, with peak flowering being, on average, later than for NSshrubs and NNshrubs.

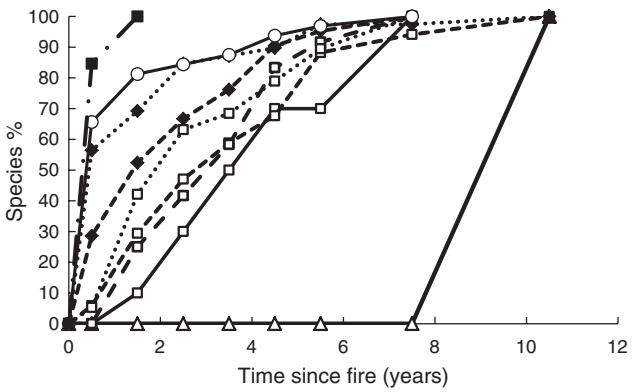


Fig. 3. The percentage of species in each plant functional type (PFT) that had reached first flowering per time since fire (TTFF) sampling year: NStree (△), NSshrub (□), NNshrub (□, long-dashed line), RSshrub (□, short-dashed line), RNshrub (□, dotted line), NNlow (◆, dashed line), RNlow (◆, dotted line), geo (○), ephem (■). See Table 1 for descriptions of each PFT.

The non-resprouting serotinous (NSshrub) species have a long TTFF (Fig. 3), with some taking 7.5 years (e.g. *Hakea gilbertii*, *Petrophile squamata* and *Allocasuarina campestris*). *Banksia armata* and *Beaufortia micrantha* took 15.5 years to reach peak flowering, compared with the NSshrub mean of 11.8 years. The non-resprouting, non-serotinous shrubs (NNshrubs) tended to flower and peak earlier than did the NSshrubs, replenishing their soil-stored seed banks. These include many species in the Fabaceae (e.g. *Acacia* spp. and *Gastrolobium* spp.) and species in Myrtaceae (e.g. *Verticordia* spp. and *Calytrix* spp.). However, ~80% of NNshrub species showed a marked decline in flowering and observations by 30 years.

Although very common, *A. huegeliana* was the only non-resprouting serotinous tree (NStree) species, and it took longer to flower (mean 10.5 years) and reach peak flowering (25 years) than the average of any other PFTs.

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

Table 4. Paired contrasts in time to first flowering (TTFF) and time to peak flowering (TTPF) among plant traits

In the planned contrasts among plants of different stature, the contrast was limited to shrubs and low layers because there were data on only a single tree. n.s., not significant. ***$P < 0.001$; **$P < 0.01$; *$P < 0.05$

Contrast	Time to first flowering				Time to peak flowering			
	Mean ± s.e.	*t*	d.f.	Significance	Mean ± s.e.	*t*	d.f.	Significance
Resprouters vs	2.13 ± 0.2	–4.48	1166	***	7.49 ± 1.0	–1.61	1,55	n.s.
non-resprouters	3.59 ± 0.3				6.15 ± 1.3			
Geophytes vs	1.44 ± 0.3	–2.69	1100	**	1.07 ± 0.6	–4.54	1,32	***
other resprouters	2.44 ± 0.3				9.15 ± 1.0			
Serotinous non-resprouters[A] vs	5.05 ± 0.9	2.18	1,64	*	14.8 ± 3.6	2.47	1,21	*
non-serotinous non-resprouters	3.30 ± 0.3				7.88 ± 1.0			
Shrubs vs	3.70 ± 0.3	6.77	1165	***	11.3 ± 0.7	8.90	1,54	***
low	1.80 ± 0.2				3.24 ± 0.6			

[A]This includes non-resprouting trees and shrubs. If the NStree is excluded, the test result is not significant.

Fire-response traits and flowering patterns

General fire-response traits successfully predicted TTFF, but predictions of TTPF in some cases were contrary to expectations (Table 4). The mean TTFF for resprouter species was significantly less than that for non-resprouting species (2.13 years vs 3.59 years, $P < 0.001$), but contrary to our prediction, mean TTPF was longer, although not significantly different between the two trait states (7.49 years vs 6.15 years, $P = 0.113$). Mean TTFF for geophytes was significantly shorter than that for other resprouters (1.44 years vs 2.44 years, $P = 0.008$) as was mean TTPF (1.07 years vs 9.15 years, $P < 0.001$). Among non-resprouting taxa, mean TTFF of serotinous species was significantly longer than that for non-serotinous species (5.05 years vs 3.30 years, $P = 0.033$), as was mean TTPF (14.8 years vs 7.88, $P = 0.022$), although the latter was contingent on inclusion of the slowly maturing tree *A. huegeliana*. A comparison excluding this species (the sole tree in our study) showed that mean TTPF was not significantly different between the two trait states. Plant stature had a substantial effect on the mean TTFF and TTPF, with shrubs taking significantly longer to first flowering (3.70 years vs 1.80 years, $P < 0.001$) and peak flowering (11.3 vs 3.24 years, $P < 0.001$) than for species in the low stratum. Although not included in the statistical test, the longer TTFF and TTPF of the sole tree (cf. shrubs and low species) is consistent with stature having a major effect on flowering patterns.

Discussion

The PFT approach for assessing trends in flowering patterns

Differences in TTFF and TTPF among fire-response trait states and PFTs occurred as predicted, providing further evidence for the utility of this framework for understanding vegetation dynamics in temperate southern Australian heaths, shrublands and woodlands (Keith *et al*. 2007; Gosper *et al*. 2012*a*, 2013*a*). As predicted, resprouters flowered sooner than did non-resprouters, which agrees with the findings of Keith *et al*. (2007) and Burrows *et al*. (2008). However, resprouting species did not reach peak flowering sooner. This seemingly inconsistent response (more rapid TTFF, but longer TTPF) is consistent with bet-hedging strategies. Resprouters may rapidly accumulate a rudimentary seed bank after fire, allowing the potential for seedling recruitment in the event of a very short fire interval to take advantage of likely lower competition with co-occurring non-resprouters, but have resource trade-offs over the medium term, delaying high investment in sexual reproduction in favour of allocating resources to ensure individual persistence (Bond and Midgley 2001). Resprouting serotinous and non-serotinous species continue to flower over an extended period post-fire. Serotinous non-resprouters (NStree and NSshrubs) had relatively long juvenile periods and continued flowering in long-unburnt vegetation, whereas non-serotinous non-resprouters (persistent soil-stored seed; NNshrubs) had a shorter average TTFF and TTPF and species presence declined after an earlier peak. This was consistent with a strategy of prioritising seed-bank accumulation in NNshrubs and NNlows before competitive suppression by dominant NSshrubs (or NStrees in woodland), and relying on this seed bank for population continuity after the next fire or alternate trigger (Keith *et al*. 2007; Gosper *et al*. 2012*a*). Our reproductive vegetation-dynamics model assumes that inter-fire recruitment in most perennial plant species is rare and that, over the time frame of the present study, any inter-fire recruits were unlikely to contribute to observed flowering responses.

Geophytes, a special class of resprouters that avoid summer and autumn fires through retreating to belowground organs, flower sooner following fire than do other resprouters, for which aboveground biomass is reduced by burning. Burrows *et al*. (2008) observed a similar trend in jarrah forest where geophytes flowered earlier after fire than did resprouting shrubs, perennial herbs, grasses and sedges. Our findings indicated the utility of separating geophytes as a special class of RNlow for interpreting and predicting the effects of fires. Prescribed fire in spring may have a considerably different effect on geophytes because the aboveground leaves and or flowers would have been burnt at a time when the belowground organs would be depleted (Lamont and Downes 2011).

Also concurring with Burrows *et al*. (2008), in our study, plants from taller strata tended to have longer TTFF and TTPF than did those in lower strata, possibly reflecting a greater allocation of resources to growth and attaining site dominance over reproduction in the years following fire. This finding suggests that stature may be a valuable trait to consider in efforts to identify species with potentially high sensitivity to short fire intervals.

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

Strengths and constraints of study

Long-term ecological research can reveal important insights into ecological processes that are difficult or impossible to detect with chronosequence or short-term experiments (Likens 1989). The present study was unique because there have been no additional fires within the 10- and 30-year longitudinal study periods. Multiple prescribed and wild fires have occurred in other longitudinal studies, for example, in alpine environments (McDougall 2007), tropical savanna woodlands (Russell-Smith *et al*. 2003), coastal heath (Freestone *et al*. 2015) and mallee (Keith and Tozer 2012).

The value of the present study lies in the investigation of the effects of TSF on plant reproduction in the same plots over a long time scale compared to other similar studies in Australia. Consequently, our results are not confounded by location, an acknowledged shortcoming of chronosequence studies. Furthermore, the Tutanning study included multiple observations soon after the fire events, capturing important information on early reproduction often not available in chronosequence studies. Nevertheless, our approach combining two longitudinal studies assumed that differences between the two fires (event effects) do not confound post-fire trends in reproduction. We have minimised potential issues by using fires that occurred at the same time of year, were of similar intensity and winter rainfall was similar in the first few years following both fires.

Management implications and determination of appropriate fire intervals

Our study has provided information on TTFF and TTPF for species and PFTs that is critical for predicting how plant communities in the western part of the Avon Wheatbelt Bioregion will respond to fire management. Having both early post-fire records and long-term records would enhance studies in semiarid shrublands that lack early post-fire data and same-site long-term records (Knuckey *et al*. 2016) The determination of appropriate minimum fire intervals for plant species are often based on the juvenile period of key fire-sensitive flora, or 'rules of thumb' applied to these data (Burrows *et al*. 2008; Gosper *et al*. 2013*a*). Twice the time it takes for the most fire-sensitive species to first flower, or twice the time when 50% of observations of that species are in flower, or time to reach peak flowering, are examples of alternative metrics to determine minimum fire intervals on the basis of the juvenile period. Our measurement of TTPF provided an additional estimate of the minimum fire interval needed for a species, so as to have sufficient seeds for population replacement.

Non-resprouting serotinous species are usually regarded as the most fire-sensitive species (Keith *et al*. 2002*a*, 2002*b*; Bradstock and Kenny 2003; Gosper *et al*. 2013*a*). In our study, *A. huegeliana* is the only NStree, which first flowers at 10 years, has 50% of plots recording flowering at some point beyond 15 years and takes at least 25 years to reach peak flowering, which would suggest minimum fire intervals for woodlands of 20–30 years for this species to be able to replace itself. Vegetation structural features that may be important for a range of fauna may take even longer to develop and should be considered in any fire planning (Cale and Burbidge 1981; Keith *et al*. 2002*a*).

Identified as a critical group for fire management in the western weatbelt, the non-resprouting serotinous shrubs (NSshrubs), mainly from the Proteaceae, Casuarinaceae and Myrtaceae families, take between 8 and 16 years to reach peak flowering. As these are dominant components of the kwongan vegetation, a working hypothesis for the minumum fire interval to reduce immaturity risk for this vegetation association could be 15–20 years. As there is a danger of not maintaing full species diversity under a regime of constant fire intevals (Keith *et al*. 2002*a*), alowing the intervals to vary within a defined fire-interval range could help avoid competitive dominance by a small subset of species and assist in maintaing overall levels of plant diversity. Retaining some long-unburnt vegetation in the landscape may also be beneficial to maximise vegetation composition and structural diversity, provide habitat for fauna, and become reference sites where ongoing flowering patterns, plant-mortality rates and inter-fire recruitment can be studied (Knuckey *et al*. 2016).

Burrows *et al*. (2008) suggested that TTFF within SWWA forest communities increases with aridity. Our study in lower-rainfall kwongan and woodland was consistent with this. Generally, juvenile periods were longer for shrub and woodland species at Tutanning than were those measured for species in the wetter forests (Burrows *et al*. 2008), but similar or slightly shorter than those in mallee-heath and mallee shrublands in a drier environment (Gosper *et al*. 2013*a*). Consequently, it is not surprising that TTFF in PFTs were substantially longer in the present study, than are their equivalents in wetter environments elsewhere in Australia, for example, in dry sclerophyll forests (Bradstock and Kenny 2003), wet heath (Clarke *et al*. 2009) and heath (Benson 1985; Keith *et al*. 2002*a*). Compared with other Australian shrublands, the appropriate minimum fire-interval range (15–20 years) suggested here for kwongan is roughly similar with the 13–30 years for fire-sensitive serotionous, obligate-seeding shrubs in species-rich heathlands near Sydney (Keith *et al*. 2002*b*), longer than the 5–20 years suggested for more mesic shrublands north of Perth (Enright *et al*. 2011), the 6–35 years for very wet heaths in the New England Tablelands (Clarke *et al*. 2009), but lower than the ~30 years suggested for more arid mallee–heath in the south-eastern wheatbelt (Gosper *et al*. 2013*a*).

The longitudinal approach used here has not sampled a sufficiently long period since fire to inform ecologically based maximum fire intervals. However, studies in the present and related ecosystems have suggested, in heavily fragmented agricultural landscapes where fire is no longer operating across the landscape, that defining maximum intervals is important and should be a priority for research and management (Yates and Ladd 2005; Maher *et al*. 2010; Gosper *et al*. 2012*b*, 2013*b*; Shackelford *et al*. 2015).

Conflicts of interest

None.

Acknowledgements

We acknowledge the Department of Biodiversity Conservation and Attractions and its predecessors for providing ongoing funding and support to undertake this research. We also thank Brett Beecham, Greg

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025

Durell and Peter Lacey in the Parks and Wildlife Wheatbelt Region for continuing support for the project. This paper is dedicated to Angas Hopkins who initiated the fire studies at Tutanning more than 35 years ago.

References

Benson DH (1985) Maturation periods for fire-sensitive shrub species in the Hawksbury sandstone vegetaion. *Cunninghamia* **1**, 339–349.

Bond WJ, Midgley JJ (2001) Ecology of sprouting in woody plants: the persistence niche. *Trends in Ecology & Evolution* **16**, 45–51. doi:10.1016/S0169-5347(00)02033-4

Bond WJ, van Wilgen BW (1996) 'Fire and plants.' (Chapman and Hall: London)

Bradstock RA, Kenny BJ (2003) An application of plant functional types to fire management in a conservation reserve in southeastern Australia. *Journal of Vegetation Science* **14**, 345–354. doi:10.1111/j.1654-1103.2003.tb02160.x

Brown JM, Hopkins AJM (1983) The kwongan (sclerophyllous shrublands) of Tutanning Nature Reserve, Western Australia. *Australian Journal of Ecology* **8**, 63–73. doi:10.1111/j.1442-9993.1983.tb01519.x

Burrows N, Wardell-Johnson GW, Ward B (2008) Post-fire juvenile period of plants in south-west Australian forests and implications for management. *Journal of the Royal Society of Western Australia* **91**, 163–174.

Cale PG, Burbidge AH (1981) 'Research plan for the western ground parrot, western whipbird and western bristlebird.' (Australian National Parks and Wildlife Service: Canberra)

Clarke PJ, Knox KJE, Campbell ML, Copeland LM (2009) Post fire recovery of woody plants in the New England Tableland Bioregion. *Cunninghamia* **11**, 221–239.

Cramer MD, Midgley JJ (2009) Maintenance costs of serotiny do not explain weak serotiny. *Austral Ecology* **34**, 653–662. doi:10.1111/j.1442-9993.2009.01971.x

Egler FE (1954) Vegetation science concepts. I. Initial floristic composition, a factor in old-field vegetation development. *Vegetatio* **4**, 412–417. doi:10.1007/BF00275587

Enright NJ, Fontaine JB, Westcott VC, Lade JC, Miller BP (2011) Fire interval effects on persistence of resprouter species in Mediterranean-type shrublands. *Plant Ecology* **212**, 2071–2083. doi:10.1007/s11258-011-9970-7

Freestone M, Wills TJ, Read J (2015) Post-fire succession during the long-term absence of fire in coastal heathland and a test of the chronosequence survey method. *Australian Journal of Botany* **63**, 572–580. doi:10.1071/BT14345

Gosper CR, Yates CJ, Prober SM (2012*a*) Changes in plant species and functional composition with time since fire in two Mediterranean climate plant communites. *Journal of Vegetation Science* **23**, 1071–1081. doi:10.1111/j.1654-1103.2012.01434.x

Gosper CR, Yates CJ, Prober SM, Parsons BC (2012*b*) Contrasting changes in vegetation structure and diversity with time since fire in two Australian Mediterranean-climate plant communities. *Austral Ecology* **37**, 164–174. doi:10.1111/j.1442-9993.2011.02259.x

Gosper CR, Prober SM, Yates CJ (2013*a*) Estimating fire interval bounds using vital attributes: implications of uncertainlty and among population variabilty. *Ecological Applications* **23**, 924–935. doi:10.1890/12-0621.1

Gosper CR, Prober SM, Yates CJ (2013*b*) Multi-century changes in vegetation structure and fuel availability in fire sensitive woodlands. *Forest Ecology and Management* **310**, 102–109. doi:10.1016/j.foreco.2013.08.005

Keeley JE, Pausas JG, Rundel PW, Bond WJ, Bradstock RA (2011) Fire as an evolutionary pressure shaping plant traits. *Trends in Plant Science* **16**, 406–411. doi:10.1016/j.tplants.2011.04.002

Keith DA, Bradstock RA (1994) Fire and competition in Australian heath: a conceptual model and field investigations. *Journal of Vegetation Science* **5**, 347–354. doi:10.2307/3235858

Keith DA, Tozer MG (2012) The influence of fire, herbivores and rainfall on vegetation dynamic in the mallee: a long term experiment. *Proceedings of the Linnean Society of New South Wales* **134**, 35–54.

Keith DA, McCaw WL, Whelan RJ (2002*a*) Fire regimes in Australian heathlands and their effects on plants and animals. In 'Flammable Australia: the fire regimes and biodiversity of a continent'. (Eds RA Bradstock, JE Williams, AM Gill) pp. 199–237. (Cambridge University Press: Cambridge, UK)

Keith DA, Williams JE, Woinarski JC (2002*b*) Fire management and biodiversity conservation: key approaches and principles. In 'Flammable Australia: the fire regimes and biodiversity of a continent'. (Eds RA Bradstock, JE Williams, AM Gill) pp. 401–425. (Cambridge University Press: Cambridge, UK)

Keith DA, Lisa H, Suzette R, Jedda L, Michael B (2007) Plant functional types can predict decade-scale changes in fire-prone vegetation. *Journal of Ecology* **95**, 1324–1337. doi:10.1111/j.1365-2745.2007.01302.x

Knuckey CG, van Etten EJB, Doherty TS (2016) Effects of long-term fire exclusion and frequent fire on plant community composition: a case study from semi-arid shrublands. *Austral Ecology* **41**, 964–975. doi:10.1111/aec.12388

Lamont BB, Downes KS (2011) Fire-stimulated flowering among resprouters and geophytes in Australia and South Africa. *Plant Ecology* **212**, 2111–2125. doi:10.1007/s11258-011-9987-y

Likens GE (1989) 'Long term studies in ecology: approaches and alternatives.' (Springer-Verlag: New York)

Maher KA, Hobbs RJ, Yates CJ (2010) Woody shrubs and herbivory influence tree encroachment in the sandplain heathlands of southwestern Australia. *Journal of Applied Ecology* **47**, 441–450. doi:10.1111/j.1365-2664.2010.01779.x

McDougall KL (2007) Grazing and fire in two subalpine peatlands. *Australian Journal of Botany* **55**, 42–47. doi:10.1071/BT06096

Noble IR, Slayter RO (1980) The use of vital attributes to predict successional changes in plant communities subject to recurrent disturbances. *Vegetatio* **43**, 5–21. doi:10.1007/BF00121013

Parsons BC, Gosper CR (2011) Contemporary fire regimes in a fragmented and an unfragmented landscape: implications for vegetation structure and persistence of the fire-sensitive malleefowl. *International Journal of Wildland Fire* **20**, 184–194. doi:10.1071/WF09099

Russell-Smith J, Whitehead PJ, Cook GD, James LH (2003) Response of eucalyptus-dominated savanna to frequent fires: lessons from Munmarlary, 1973–1996. *Ecological Monographs* **73**, 349–375. doi:10.1890/01-4021

Shackelford N, Renton M, Perring MP, Brooks K, Hobbs RJ (2015) Biodiversity change in heathland and its relationships with shifting local fire regimes and native species expansion. *Journal of Plant Ecology* **8**, 17–29. doi:10.1093/jpe/rtu005

Whelan RJ (1995) 'The ecology of fire.' (Cambridge University Press: Cambridge, UK)

Yates CJ, Ladd PG (2005) Relative importance of reproductive biology and establishment ecology for persistence of a rare shrub in a fragmented landscape. *Conservation Biology* **19**, 239–249. doi:10.1111/j.1523-1739.2005.00286.x

Handling Editor: James Camac

Downloaded from http://connectsci.au/bt/article-pdf/65/4/339/831064/bt17011.pdf by UNSW Library user on 09 December 2025